

CIA Triad



Estimated time needed: 5 minutes

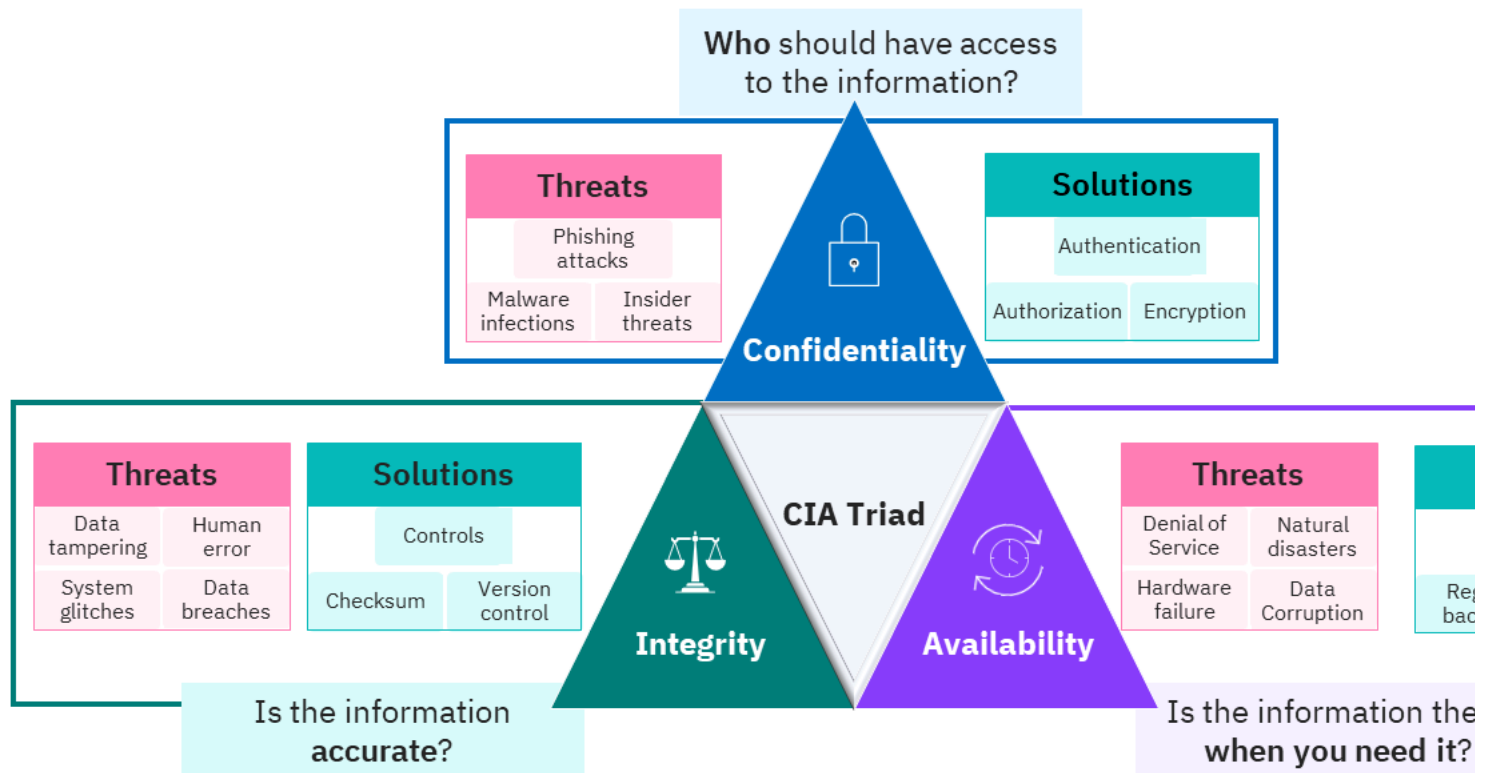
Objectives

- Summarize the CIA Triad
- Review the different solutions to ensure confidentiality, integrity, and availability

CIA Triad: The Cybersecurity Cornerstone

The CIA Triad serves as the cornerstone of the cybersecurity field, guiding every aspect, from the design of secure systems to the implementation of security policies and procedures. The graphic provides an overview of the following questions:

- What is it?
- How can it be breached?
- How can you prevent an attack?



Solutions to Ensure Confidentiality

- **Authentication:** Passwords, PINs, and multi-factor authentication methods are used to verify the identity of the user
- **Authorization:** Verify whether the authenticated user should have access to the requested information
- **Encryption:** Transform readable data into a coded form that prevents unauthorized access to the information

Solutions to Ensure Integrity

- **Controls:** Help to protect data from unauthorized creation, deletion, or alterations
- **Checksum:** Verify data integrity by detecting errors that may have been introduced during transmission or storage
- **Version control:** Record changes to a file over time so that you can see who and when changes were made to the file

Solutions to Ensure Availability

- **Fail-safe mechanisms:** Automatically respond to a failure or malfunction to prevent further damage or harm
- **Perform regular backups:** Routinely create copies of data and store them in a separate location to restore the original data in the event of data loss or disruption
- **Disaster recovery plan:** Create policies to ensure the rapid restoration of systems, data, and infrastructure in the event of natural disasters, cyber-attacks, and other unforeseen events

Author

Dee Dee Collette



Skills Network